

The Chinese University of Hong Kong, Shenzhen

Course Outline

1. Course identity

A. Course as listed at CUHK-Shenzhen

The information in this block should be exactly as approved by CUHK Senate. In case there are any differences, please explain in the table below.

Course code	STA5013
Course title (English)	Causal Inference
Course title (Chinese)	因果推理
Units	3
Description (English)	Investigating causality between events is an important topic in modern-day statistics. This class introduces basic concepts, methods, and theoretical results on causal inference. Topics include: 1) Randomization experiment: Potential outcomes, Complete randomization experiment and Fisher Randomization Test, Stratification, Regression Adjustment. 2) Observational study: Selection bias, Propensity score and Inverse propensity score weighting estimator, Doubly Robust estimation, E-Value. 3) Instrumental Variable: Definition, Linear IV model, Fuzzy Regression Discontinuity.
Description (Chinese)	发现事件间的因果关系是现代统计学的一个基础问题。本课程旨在介绍因果推断的基本概念及方法。课程包括：1.随机化试验：潜在结果模型，完全随机实验与费雪随机化检验，分层。2.观察性研究：倾向得分及逆概率加权估计，双稳健估计，E-值。3.工具变量

B. Corresponding course at CUHK

Please give details of the closest corresponding course at CUHK (as approved by CUHK Senate and listed in course list). If the course in Shenzhen maps to more than one course at CUHK, please make multiple copies of the block below.

Course code	N/A
Course title (English)	
Course title (Chinese)	
Units	
Description (English)	

Description (Chinese)	
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2. Course requisites

Please state prerequisites and co-requisites, in terms of courses at CUHK-Shenzhen or any other requirements.

A. Prerequisites

STA2002H Honours Probability and Statistics or STA2002 Probability and Statistics II or having permission from academic advisor.

B. Co-requisites

C. Exclusion

D. Others

3. Learning outcomes

1. Understand basic concepts and methods in causal inference discipline.
2. Understand how to perform randomization experiments for real-life data.
3. Know how to perform observational study.

4. Course syllabus

Please highlight fundamental concepts involved in each topic to help students better understand what is covered in the course.

The goal of this course is to introduce the concepts, methods, and theories regarding the causal inference discipline. We will provide written homework & programming assignments to help achieve this goal. Topics covered will include the following:

1. Review & Introduction
 - 1.1 The history of causal inference
 - 1.2 Hypothesis testing, Linear regression, and Logistic regression
2. Randomization experiment
 - 2.1 Potential outcome and Yule-Simpson paradox
 - 2.2 Complete randomization experiment and Fisher Randomization Test
 - 2.3 Stratification in randomization experiment

2.4 Regression adjustment

3. Observational study

3.1 Selection bias and model identification

2.2 Propensity score and IPW estimator

2.3 Doubly Robust estimation

2.4 E-value

4. Instrumental Variable

4.1 Definition and its application in economy

4.2 Linear IV model

4.3 Fuzzy Regression Discontinuity Design

5. Assessment scheme

Component/ method	% weight
Homework Assignments	30
Midterm Exam	35
Final Project	35

Remarks:

6. Grade descriptor

The grade descriptor should align with specific course learning outcomes, not a generic description applicable to all courses.

Type: **General**

Grade	Description
A/A-	EXCELLENT - exceptionally good performance and exceeding expectations in all or most of the course learning outcomes; demonstration of a solid understanding of the subject matter, the ability to analyze problems and apply extensive knowledge, and skillful use of concepts and materials to derive proper solutions.
B+/B/B-	GOOD – good performance in all course learning outcomes and exceeding expectations in some of them; demonstration of a good understanding of the subject matter and the ability to use proper concepts and materials to solve most of the problems encountered.
C+/C/C-	FAIR – adequate performance and meeting essential expectations in course learning outcomes; demonstration of adequate understanding of the subject matter and the ability to solve simple problems.
D+/D	MARGINAL – performance barely meets expectations in the course learning outcomes; demonstration of partial understanding of the subject matter and the ability to solve simple problems.
F	FAILURE – performance does not meet expectations in the essential course learning

	outcomes; demonstration of serious deficiencies and the need to retake the course.
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7. Feedback for evaluation

Please specify forms of evaluation to generate feedback on classes from students.

Informal feedback to the instructor and/or teaching assistant(s) through face-to-face talking, emails, office hours, etc.;

Advisor-advisee program and program review.

8. Readings

Required:

1.A First Course in Causal Inference, 978-1-032-75862-6, Peng Ding, Chapman and Hall/CRC, 1, 2024, United States

Others:

9. Course components

Activity	Hours/week
Assignments	4
Lecture	3

10. Indicative teaching plan

Week	Content/ topic/ activity
3	Yule-Simpson paradox, Complete randomization experiment and Fisher Randomization Test
4	Neymanian repeated sampling inference
5	Stratification in randomization experiment and Regression adjustment
6	Selection bias and model identification in observational study, Propensity score and its estimation
7	IPW estimator and parametric model
8	Review and midterm
9	Doubly Robust estimation
10	E-value

11	Definition of instrumental variable and its application in economy, Linear IV model
1-2	Introduction, Review of Hypothesis testing, Linear regression, Logistic regression, Potential outcome framework
12	Local linear regression and Fuzzy Regression Discontinuity Design
13-14	Final Project

11. Implementation plan

The implementation plan may vary from year to year. Please indicate expected enrolment, and number of sections. Example: 150 students for lecture (x 2); 30 students for tutorials (x 10).

About 40 people a class and there are 1 session in total

12. Academic honesty and plagiarism

A course outline may include subject-specific requirements on plagiarism, in addition to the University established policies.

Students must complete all coursework and assessments independently, unless AI tool use is explicitly permitted. Unauthorized or improper use of AI may constitute academic dishonesty and harm learning. Misuse includes submitting AI-generated work as one's own, using AI to cheat, or employing AI unethically. Always obtain permission before using AI in academic work, and be sure to acknowledge any AI use in submissions.

13. Approval

Has the course title been included in the programme submission approved by CUHK Senate? Are there any differences? Have the details (as in this document) been approved at School or other level in CUHK- Shenzhen?

Submit application

14. Any other information

15. Version date

Version number	1
As of (date)	2025-10-31